

# eZ430-Chronos Custom Firmware - Benutzerhandbuch

Custom Beta - MSP430-GCC - CC430F6137

Dieses Handbuch beschreibt den aktuellen Beta-Stand der angepassten MSP430-GCC-Firmware.

## Tasten / Buttons / Mga Pindutan

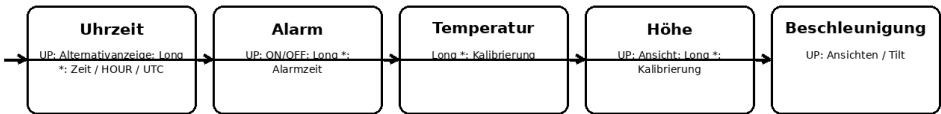
| Button     | Function                                                                 |
|------------|--------------------------------------------------------------------------|
| * kurz     | Oberes Menü weiterschalten.                                              |
| * lang     | Einstellungen des aktuellen oberen Menüpunktes öffnen.                   |
| # kurz     | Unteres Menü weiterschalten.                                             |
| # lang     | Einstellungen des aktuellen unteren Menüpunktes öffnen.                  |
| UP         | Direktfunktion oben; im Alarm-Menü Alarm EIN/AUS.                        |
| DOWN       | Direktfunktion unten.                                                    |
| LIGHT kurz | Beleuchtung einschalten.                                                 |
| LIGHT lang | Lichtmenü: DOWN 1-30 s, UP Dauerlicht ON/OFF, LIGHT speichern/verlassen. |

## Menu

LINE 1 (\*): Uhrzeit -> Alarm -> Temperatur -> Höhe -> Beschleunigung

LINE 2 (#): Datum -> TZ2 -> Swatch Beat Time -> BIN Watch -> Stoppuhr -> Countdown -> Random -> Batterie

### Menüstruktur - obere Zeile (\*)



#### Bedienung

\* kurz: nächster oberer Menüpunkt  
UP: Direkt-/Alternativfunktion  
\* lang: Einstellmenü

Figure A - Line 1 / \*

## Menüstruktur - untere Zeile (#)

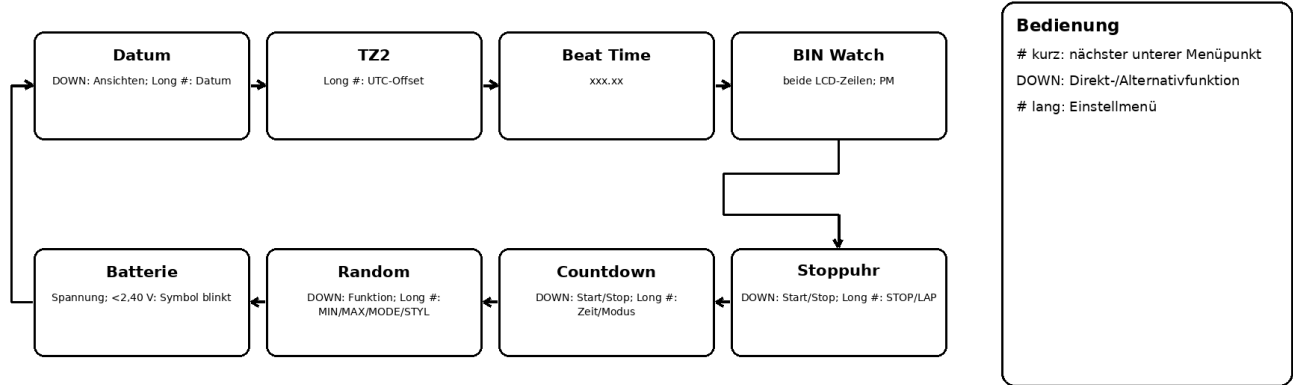


Figure B - Line 2 / #

## Funktionen / Functions / Mga Function

### Uhrzeit

Longpress \* öffnet die Uhreinstellungen. HOUR schaltet das stündliche Doppel-Piep ein/aus. UTC definiert den Abstand der angezeigten Ortszeit zu UTC; DST/Sommerzeit wird zusätzlich berücksichtigt.

### Alarm

UP schaltet EIN/AUS und zeigt kurz ON/OFF. Longpress \* stellt die Alarmzeit ein.

### Temperatur

Interne Temperaturanzeige; Longpress \* öffnet die vorhandene Einstellung/Kalibrierung.

### Höhe

Höhen-/Luftdruckfunktion des Drucksensors; vorhandene Direkt- und Einstellfunktionen bleiben erhalten.

### Beschleunigung

Beschleunigungs-/Tilt-Anzeige des Sensors.

### Datum

Datumsansichten; Longpress # öffnet die Datumseinstellungen.

### TZ2

Zweite Zeitzone. Longpress # stellt den Offset ein. Halbstundenzonen werden unterstützt, z. B. Indien +5,5 h. Anzeige mit Stunden, Minuten und Sekunden.

### Swatch Beat Time

Beat-Anzeige im Format xxx.xx, aus UTC berechnet. Benutzer-UTC und DST werden berücksichtigt.

### BIN Watch

Binäres Watchface über beide LCD-Zeilen; PM kennzeichnet die zweite Tageshälfte.

## Stoppuhr

DOWN bedient die Direktfunktion; Longpress # öffnet die Moduseinstellung STOP/LAP.

## Countdown

Countdown-Timer mit Direktfunktion und Einstellmenü.

## Random

Zufallszahlengenerator mit MIN, MAX, MODE und STYL. MIN=1/MAX=6 erzeugt nur 1 bis 6.

## Batterie

Beim Öffnen erfolgt eine ADC-Messung; während der Anzeige wird aktualisiert. Unter 2,40 V blinkt das Batteriesymbol global. Die absolute Spannung im aktuellen Beta-Stand als Trendwert betrachten.

## Beta / Technical

Watchdog ist aktiv. LPM3/Sleep ist absichtlich deaktiviert, weil der frühere Sleep-Pfad nach längerer Laufzeit fehlerhafte bzw. verzögerte Tastenreaktionen verursachte.

Build: make clean && make FREQUENCY=868 SUPPORT\_DIR=C:/ti/msp430-gcc/include

# eZ430-Chronos Custom Firmware - User Manual

## Custom Beta - MSP430-GCC - CC430F6137

This manual describes the current beta build of the customized MSP430-GCC firmware.

### Tasten / Buttons / Mga Pindutan

| Button      | Function                                                         |
|-------------|------------------------------------------------------------------|
| short *     | Advance upper menu.                                              |
| long *      | Open settings for current upper item.                            |
| short #     | Advance lower menu.                                              |
| long #      | Open settings for current lower item.                            |
| UP          | Upper direct action; in Alarm toggles ON/OFF.                    |
| DOWN        | Lower direct action.                                             |
| short LIGHT | Turn on backlight.                                               |
| long LIGHT  | Light setup: DOWN 1-30 s, UP continuous ON/OFF, LIGHT save/exit. |

### Menu

LINE 1 (\*): Time -> Alarm -> Temperature -> Altitude -> Acceleration

LINE 2 (#): Date -> TZ2 -> Swatch Beat Time -> BIN Watch -> Stopwatch -> Countdown -> Random -> Battery

#### Menu structure - upper line (\*)

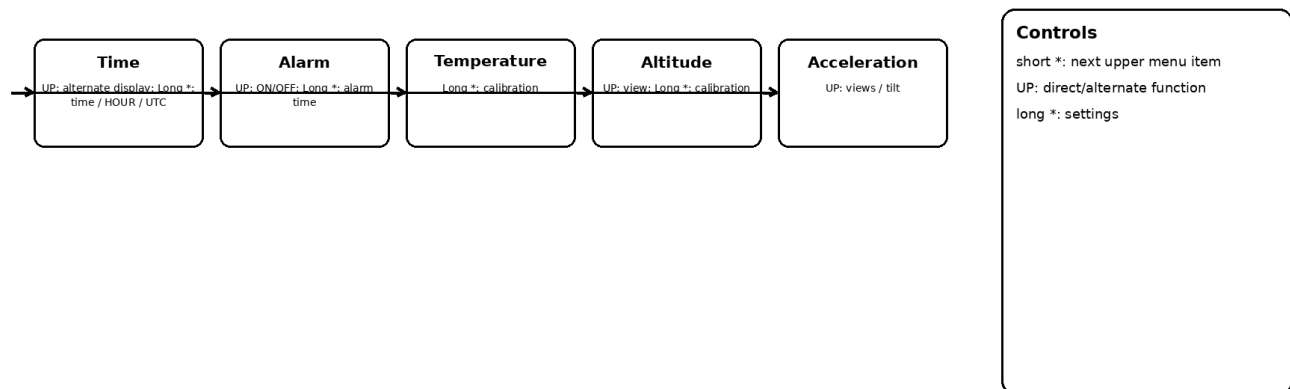


Figure A - Line 1 / \*

## Menu structure - lower line (#)

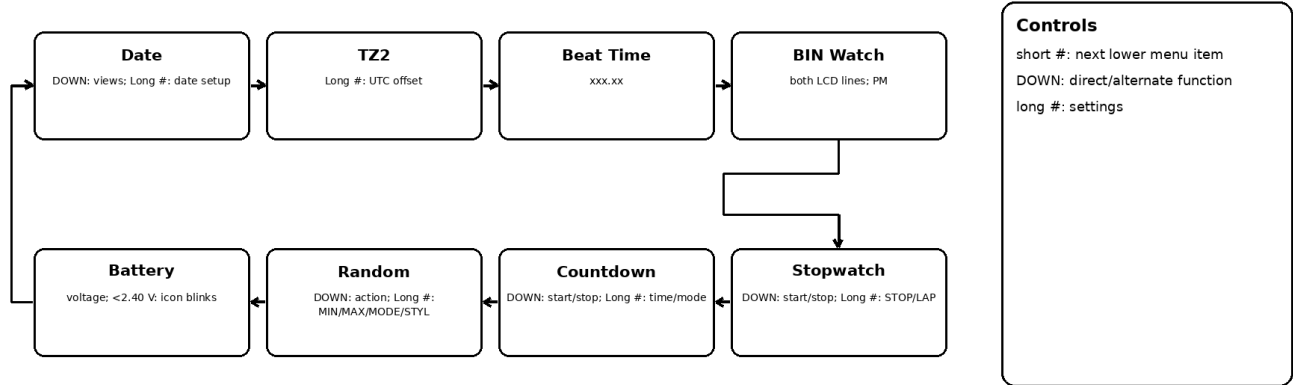


Figure B - Line 2 / #

## Funktionen / Functions / Mga Function

### Time

Long-press \* opens time settings. HOUR toggles the hourly double beep. UTC defines the user-selected relationship between displayed local time and UTC; DST is handled separately.

### Alarm

UP toggles ON/OFF and briefly displays ON/OFF. Long-press \* sets alarm time.

### Temperature

Internal temperature display; long-press \* opens existing setup/calibration.

### Altitude

Altitude/pressure sensor function; existing direct/setup functions are retained.

### Acceleration

Accelerometer/tilt display.

### Date

Date views; long-press # opens date settings.

### TZ2

Second time zone. Long-press # sets the offset. Half-hour zones are supported, e.g. India +5.5 h. Displays hours, minutes and seconds.

### Swatch Beat Time

Beat display in xxx.xx format, derived from UTC using the configured UTC relationship and DST.

### BIN Watch

Binary watch face using both LCD lines; PM marks the second half of the day.

## Stopwatch

DOWN operates the direct function; long-press # opens STOP/LAP mode setup.

## Countdown

Countdown timer with direct control and settings.

## Random

Random generator with MIN, MAX, MODE and STYL. MIN=1/MAX=6 produces only 1 through 6.

## Battery

A real ADC measurement is triggered on entry and refreshed while visible. Below 2.40 V the battery icon blinks globally. In this beta, treat absolute voltage primarily as a trend value.

## Beta / Technical

Watchdog is enabled. LPM3/sleep is intentionally disabled because the previous sleep path caused delayed or incorrect button behavior after longer runtime.

Build: make clean && make FREQUENCY=868 SUPPORT\_DIR=C:/ti/msp430-gcc/include

# eZ430-Chronos Custom Firmware - Manwal ng Gumagamit

## Custom Beta - MSP430-GCC - CC430F6137

Inilalarawan ng manwal na ito ang kasalukuyang beta build ng binagong MSP430-GCC firmware.

### Tasten / Buttons / Mga Pindutan

| Button         | Function                                                          |
|----------------|-------------------------------------------------------------------|
| maikling *     | Lumipat sa susunod na upper menu.                                 |
| mahabang *     | Buksan ang settings ng kasalukuyang upper item.                   |
| maikling #     | Lumipat sa susunod na lower menu.                                 |
| mahabang #     | Buksan ang settings ng kasalukuyang lower item.                   |
| UP             | Direct action sa itaas; sa Alarm, ON/OFF.                         |
| DOWN           | Direct action sa ibaba.                                           |
| maikling LIGHT | Buksan ang backlight.                                             |
| mahabang LIGHT | Light setup: DOWN 1-30 s, UP tuloy-tuloy ON/OFF, LIGHT save/exit. |

### Menu

LINE 1 (\*): Oras -> Alarm -> Temperatura -> Altitude -> Acceleration

LINE 2 (#): Petsa -> TZ2 -> Swatch Beat Time -> BIN Watch -> Stopwatch -> Countdown -> Random -> Baterya

#### Istruktura ng menu - upper line (\*)

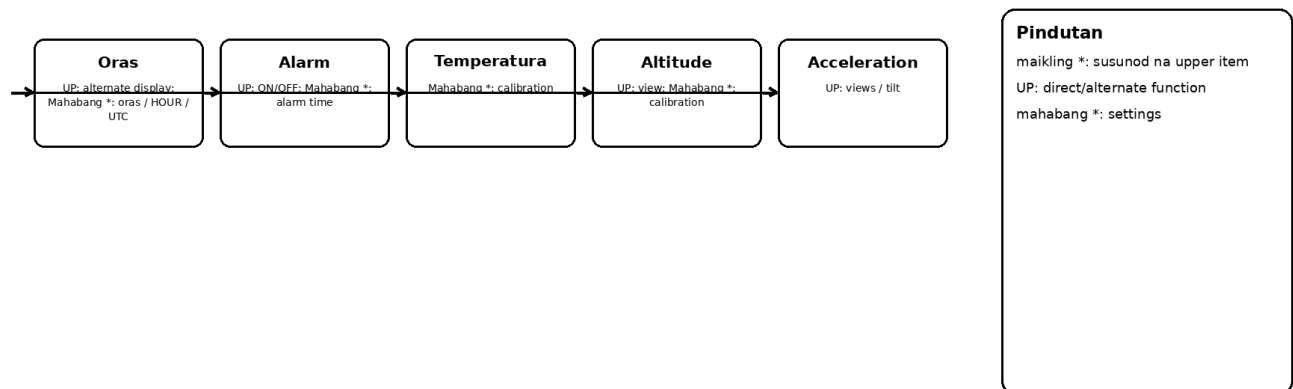


Figure A - Line 1 / \*

## Istruktura ng menu - lower line (#)

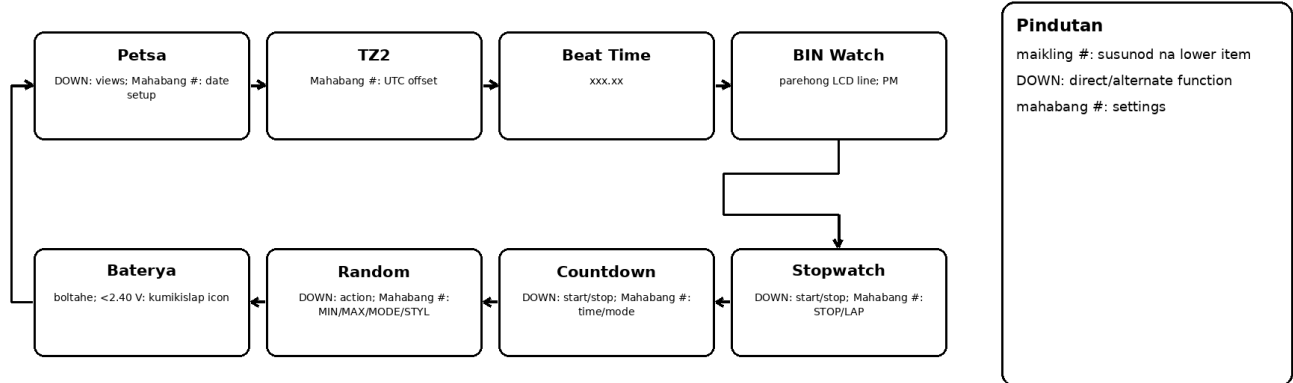


Figure B - Line 2 / #

## Funktionen / Functions / Mga Function

### Oras

Mahabang \* para sa time settings. HOUR ang ON/OFF ng dobleng beep bawat buong oras. UTC ang itinakdang pagitan ng oras sa display at UTC; hiwalay na isinasaalang-alang ang DST.

### Alarm

UP para ON/OFF at pansamantalang ipinapakita ang ON/OFF. Mahabang \* para itakda ang alarm time.

### Temperatura

Internal temperature display; mahabang \* para sa kasalukuyang setup/calibration.

### Altitude

Altitude/pressure function ng pressure sensor; nananatili ang kasalukuyang direct/setup functions.

### Acceleration

Acceleration/tilt display ng accelerometer.

### Petsa

Mga date view; mahabang # para sa date settings.

### TZ2

Ikalawang time zone. Mahabang # para sa offset. Suportado ang half-hour zones, hal. India +5.5 h. May oras, minuto at segundo.

### Swatch Beat Time

Beat display na xxx.xx, kinukuwenta mula sa UTC gamit ang naka-set na UTC relationship at DST.

### BIN Watch

Binary watch face gamit ang parehong LCD line; PM ang tanda ng ikalawang kalahati ng araw.



## Stopwatch

DOWN para sa direct function; mahabang # para sa STOP/LAP mode setup.

## Countdown

Countdown timer na may direct control at settings.

## Random

Random generator na may MIN, MAX, MODE at STYL. MIN=1/MAX=6 ay nagbibigay lamang ng 1 hanggang 6.

## Bateria

May tunay na ADC measurement sa pagpasok at nagre-refresh habang nakikita. Sa ibaba 2.40 V, kumikislap ang battery icon sa lahat ng menu. Sa beta na ito, gamitin ang boltahe lalo na bilang trend value.

## Beta / Technical

Naka-enable ang watchdog. Sadyang naka-disable ang LPM3/sleep dahil ang dating sleep path ay nagdulot ng delayed o maling button behavior matapos ang mahabang paggamit.

Build: make clean && make FREQUENCY=868 SUPPORT\_DIR=C:/ti/msp430-gcc/include